RAG Assignment: Baseline Retrieval vs. Cross-Encoder Re-Ranking

Course: Large Language Models & Multimodal Applications (LMA)
Deliverable: Technical Report (Concise 4-Page Submission)
Code Repository: https://github.com/adityaND19/LMA-Assignment-1
Target Corpus: Regulation (EU) 2016/679 (GDPR), Official Journal L 119 (88 pages, 444 sentence-aligned chunks)
Models: Bi-Encoder: `sentence-transformers/all-MiniLM-L6-v2` | Cross-Encoder: `cross-encoder/ms-marco-MiniLM-L-6-v2` | Generator: Google Gemini `gemini-3.1-flash-lite`

1. Architecture Comparison: Bi-Encoder vs. Cross-Encoder

The core architectural difference between a Bi-Encoder (Pipeline 1) and a Cross-Encoder (Pipeline 2) lies in **how and when token interactions are computed**.

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BI-ENCODER (Dual-Tower, Independent Embeddings):
  Query q   --> [ Transformer Enc ] --> u = MeanPool(H_q) ---\
                                                             Cosine Sim: (u . v) / (||u|| ||v||)
  Passage d --> [ Transformer Enc ] --> v = MeanPool(H_d) ---/
  * Offline: All passages d_i pre-embedded into vector space.
  * Query: Query encoded once. FAISS IndexFlatIP executes an exact linear scan O(N * d).

CROSS-ENCODER (Single-Tower, Joint Token Self-Attention):
  [CLS] + Query q + [SEP] + Passage d --> [ 6 Sequential Transformer Layers ] --> Scalar Score s(q,d)
  * Online: Full cross-attention across all tokens within every layer. No pre-computation possible.
```

1.1 Bi-Encoder (Independent Dual-Tower Representation)

- **Mechanism:** The query q and document chunk d are encoded through independent forward passes: `u = MeanPool(Encoder(q))`, `v = MeanPool(Encoder(d))`. Semantic similarity is computed via cosine similarity: `sim(q, d) = (u · v) / (||u||₂ · ||v||₂)`.
- **Advantages & Scalability:** Highly scalable. Document vectors `v_i` are computed once offline. During retrieval, the query is encoded once, and FAISS `IndexFlatIP` performs an exact brute-force inner-product scan across all N vectors in `O(N · d)` time (measured at 20.4 ms for 444 chunks on CPU). In large deployments, approximate nearest neighbour (ANN) indexes such as HNSW or IVF reduce candidate lookup to roughly logarithmic time, about `O(log N · d)`.
- **Bottlenecks:** Compressing 100–300 words into a single 384-dimensional vector creates an information bottleneck, and the query cannot influence how passage tokens are weighted. As a result, dense bi-encoders can struggle with:
  - *Lexical mismatch and terminology gaps* (e.g. query "subcontractor" vs. statutory text "another processor").
  - *Negations and qualifiers*: "SHALL NOT engage" and "SHALL engage" share almost all of their words, so their vectors can end up close together. In this experiment, however, most negation queries were still retrieved correctly (Q06, Q13); the negation failure observed (Q05) came from how the question was framed (Section 2.3).
  - *Distractor recitals* that repeat high-level keywords without containing operative rules.

1.2 Cross-Encoder (Joint Token Self-Attention)

- **Mechanism:** The query and passage are concatenated into a single input sequence: `x = [CLS], q_1, ..., q_m, [SEP], d_1, ..., d_n, [SEP]`. The full token sequence passes sequentially through all 6 transformer encoder layers. Within each layer, every token computes scaled dot-product self-attention with every other token: `Attention(Q, K, V) = softmax(Q Kᵀ / √dₖ) V`. The pooled representation of the `[CLS]` token is linearly projected to a scalar relevance score `s(q, d) ∈ ℝ`.
- **Advantages:** It captures fine-grained word-level dependencies, modifier bindings and conditional clauses. This helps it recognise whether a prohibition applies to a "processor" or a "controller", or whether an action requires "prior written authorisation".

2. Qualitative & Rank Shift Analysis

2.1 Failure Mode Categories & Rank Shift Summary

We evaluated 16 targeted GDPR stress queries across 5 failure categories. The candidate pool consists of the Top-10 dense chunks retrieved in Stage 1.

Query ID	Category	Query Text	Critical Chunk (Statutory Basis)	Dense Rank	CE Rank	Retrieval Outcome
Q01	Keyword Trap	Can companies share user passwords with third parties?	Article 4 (C187)	#5	**#1**	**RE-RANK RESCUE** (retrieval only, see 2.3)

Query ID	Category	Query Text	Critical Chunk (Statutory Basis)	Dense Rank	CE Rank	Retrieval Outcome
Q02	Role Limit	Can a data processor hire another subcontractor without permission?	Article 28(2) (C259)	#4	**#1**	**RE-RANK RESCUE**
Q03	Exception	Is processing sensitive data ever allowed without explicit consent?	Article 9 (C207–C210)	>#10	—	Both miss (ceiling)
Q04	Exception	Can a controller transfer personal data outside the EU?	Article 44 (C311)	#5	#5	Both miss (in Top 10)
Q05	Negation	What is explicitly prohibited when processing children data online?	Article 8 (C205)	#5	#5	Both miss (in Top 10)
Q06	Negation	When is automated decision-making NOT allowed?	Article 22 (C246)	#1	#2	Both hit
Q07	Buried Chunk	What is data protection by design and by default?	Article 25 (C253)	#5	**#1**	**RE-RANK RESCUE**
Q08	Role Limit	Who must appoint a Data Protection Officer?	Article 37 (C287)	#1	#1	Both hit
Q09	Buried Chunk	What information must be provided when a person requests access to their data?	Article 15 (C232)	#4	**#3**	**RE-RANK RESCUE**
Q10	Buried Chunk	What penalties exist for serious GDPR breaches?	Article 83 (C415–C417)	>#10	—	Both miss (ceiling)
Q11	Keyword Trap	How long can companies keep customer data?	Article 5 (C196)	#3	#3	Both hit
Q12	Exception	Can personal data be processed for scientific research without consent?	Article 89 (C429)	>#10	—	Both miss (ceiling)
Q13	Negation	What rights can be restricted under national law?	Article 23 (C248)	#1	#1	Both hit
Q14	Role Limit	What breach notification obligations apply to processors?	Article 33 (C273)	#1	#2	Both hit
Q15	Buried Chunk	Under what conditions can an authority impose a temporary ban?	Article 58 (C354)	#1	#2	Both hit
Q16	Keyword Trap	Do users have the right to delete their account data?	Article 17 (C236)	#3	#2	Both hit

Headline Results:

- **Baseline Top-3 Hit Rate:** 7/16 (43.8%) (Q06, Q08, Q11, Q13, Q14, Q15, Q16).
- **Re-Ranking Top-3 Hit Rate:** 11/16 (68.8%) (the 7 baseline hits plus 4 rescues: Q01, Q02, Q07, Q09).
- **Performance Gain:** +4 queries (+25.0 percentage points) in Top-3 retrieval hit rate.
- **Note:** Q01 is a rescue at the retrieval level only. Its critical chunk contains definitions rather than an answer, and both LLM answers were refusals (Section 2.3). Counting only rescues that changed the answer (Q02, Q07, Q09), re-ranking reaches 10/16 (62.5%), a gain of +18.8 percentage points.

2.2 Deep-Dive: Failure Queries and Answer 1 vs. Answer 2 Comparison

Failure Query 1: Q02 (Role Limits & Subcontracting)

- **Query:** "Can a data processor hire another subcontractor without permission?"
- **Rank Shift:** C259 (Article 28(2) Processor Obligations): **Dense Rank #4 (score: 0.531) → Cross-Encoder Rank #1 (score: +0.59)**
- *Baseline Top 3:* C265 (Article 29: Processing under authority), C127 (Recital 127: Supervisory cooperation), C033 (Recital 33: Scientific research consent). All 3 missed the operative rule.
- *Re-ranked Top 3:* C259 (Article 28(2) Sub-processor authorisation), C114 (Recital 114), C262 (Article 28(4) Sub-processor liability).
- **Answer 1 (Baseline):** "The provided context does not answer this question."
  - *Analysis:* **Omission / refusal.** Because the operative rule sat at rank #4, the baseline prompt contained only adjacent articles and background recitals. Following the prompt's constraints, the model stated that the context lacked the answer.
- **Answer 2 (Re-ranked):** "The processor SHALL NOT engage another processor without prior specific or general written authorisation of the controller [C259]. In the case of general written authorisation, the processor SHALL inform the controller of any intended changes concerning the addition or replacement of other processors, thereby giving the controller the opportunity to object to such changes [C259]."
  - *Analysis:* **Accurate and grounded in the text.** The cross-encoder promoted C259 from #4 to #1, so the LLM could cite the prohibition and its condition directly.

Failure Query 2: Q07 (Buried Chunk & Core Regulatory Definitions)

- **Query:** "What is data protection by design and by default?"
- **Rank Shift:** C253 (Article 25 Data Protection by Design and Default): **Dense Rank #5 (score: 0.589) → Cross-Encoder Rank #1 (score: +8.13)**
- *Baseline Top 3:* C085 (Recital 85), C091 (Recital 91: DPIA considerations), C323 (Article 47: Binding Corporate Rules).
- *Re-ranked Top 3:* C253 (Article 25), C085 (Recital 85), C084 (Recital 84).
- **Answer 1 (Baseline):** "The provided context does not answer this question."
  - *Analysis:* **Information miss.** Dense retrieval matched general recitals discussing data protection and did not deliver the operative definition.
- **Answer 2 (Re-ranked):** "Data protection by design and by default requires the controller to implement appropriate technical and organisational measures, such as pseudonymisation, to effectively integrate data-protection principles like data minimisation into the processing [C253]. This involves ensuring that, by default, only personal data necessary for each specific purpose of the processing are processed [C253]..."
  - *Analysis:* **Complete and precise.** The cross-encoder promoted C253 from #5 to #1, so the answer covers both data protection by design (Article 25(1)) and by default (Article 25(2)).

Failure Query 3: Q09 (Buried Chunk & Disclosure Checklist)

- **Query:** "What information must be provided when a person requests access to their data?"
- **Rank Shift:** C232 (Article 15 Right of Access): **Dense Rank #4 (score: 0.598) → Cross-Encoder Rank #3 (score: +2.48)**
- **Answer 1 (Baseline):** Stated that the context did not specify the required content, citing only ancillary text from C066, a recital about requests involving large quantities of data.
- **Answer 2 (Re-ranked):** Directly provided the core disclosure checklist under Article 15(1) (C232): confirmation of processing, access to the data, purposes of processing, categories of personal data, recipients, and envisaged storage periods.

2.3 Document-Specific Failure Modes & Ceiling Limitation

- **Keyword Traps & Definitions (Q01):** The query asks whether companies can share "user passwords". The GDPR never mentions passwords; it defines "third party" and "consent" in Article 4 (C187). The cross-encoder promoted C187 from dense #5 to CE #1 because it matches third-party disclosure, but both LLM answers correctly refused to answer, because Article 4 contains definitions rather than rules about sharing credentials. This rescue therefore improved the ranking but not the answer.
- **Negation & Constraint Framing (Q05):** Article 8 frames child consent thresholds affirmatively ("processing shall be lawful only if..."), whereas the query asks "What is explicitly prohibited...". Both models placed C205 at #5, showing that the mismatch between a negative question and positively worded text remains hard when the text contains no explicit negative words.
- **The Retrieval Ceiling Limitation (Q03, Q10, Q12):** In Q03 (Article 9 sensitive data exceptions), Q10 (Article 83 fines) and Q12 (Article 89 research), the bi-encoder ranked the critical chunks below #10 (e.g. C208 was dense #19). Because Stage 1 only passes the Top 10 candidates to Stage 2, the cross-encoder never saw these passages. **Key Finding:** *a re-ranker cannot rescue chunks outside the first-stage candidate pool, so Stage 1 recall is the upper bound for Stage 2.*

3. Latency Profiling

Retrieval and generation latencies were measured on CPU with pre-warmed models across N = 320 retrieval runs and n = 16 LLM calls using monotonic timers (`time.perf_counter()`).

Pipeline / Execution Stage	Median (ms)	Mean (ms)	p95 (ms)	Samples (n)	Overhead vs. Dense
Pipeline 1: Dense Retrieval (Query encode + FAISS IndexFlatIP Top-10)	**20.4**	17.1	21.6	320	1.0x (Baseline)
Re-Ranking Step Only (Cross-Encoder on 10 pairs + sort)	**424.0**	391.8	437.8	320	+424.0 ms
Pipeline 2: Full Two-Stage Retrieval (Dense Top-10 + Re-Ranking)	**443.8**	408.8	459.0	320	21.7x
LLM Generation: Answer 1 (Baseline Top-3 → Gemini API)	**1,351.1**	2,412.7	6,807.1	16	—
LLM Generation: Answer 2 (Re-ranked Top-3 → Gemini API)	**1,330.0**	2,269.0	7,391.3	16	—

Latency Trade-Off Analysis

1. **Retrieval vs. generation:** Re-ranking adds about 424 ms of CPU time, but end-to-end latency is dominated by LLM generation (about 1,330 ms median).
2. **Interactive feasibility:** Using medians, total round-trip time rises from about 1.37 s to about 1.77 s (+402 ms, a 29% increase). That stays under 2 seconds, in exchange for a +25.0 percentage-point gain in Top-3 retrieval hit rate.

4. Scaling Question: Why Not Run a Cross-Encoder Over the Whole Collection?

A central system design question is: *why not run a cross-encoder over the entire document corpus from the very start, and skip dense retrieval entirely?*

4.1 Asymptotic Complexity Comparison

Let N be the number of corpus chunks, L_q the query length, L_d the chunk length and d the embedding dimension.

- **Bi-Encoder Scaling:**
  - *Offline (once):* chunks are encoded once into vector space: $O(N \cdot L_d^2)$.
  - *Online (query time):* the query is encoded once: $O(L_q^2)$. Nearest-neighbour search over the pre-computed vectors needs $O(N \cdot d)$ dot products with an exact flat scan, which ANN indexing (FAISS HNSW/IVF) reduces to roughly logarithmic time: $\text{Query Complexity (Bi-Encoder)} \approx O(L_q^2 + \log N \cdot d)$.
- **Exhaustive Cross-Encoder Scaling:**
  - The query and document are concatenated: $(L_q + L_d)$ tokens.
  - Attention in each of the 6 layers is quadratic in sequence length: $O((L_q + L_d)^2)$.
  - Because cross-attention couples query and chunk tokens, **no pre-computation or indexing is possible**. Every query must evaluate all N chunks at run time: $\text{Query Complexity (Exhaustive CE)} = O(N \cdot (L_q + L_d)^2)$.

4.2 Empirical & Projected Scaling Benchmark

Using our measured CPU throughput of about 41.0 ms per (query, chunk) pair:

Collection Scale	Chunk Count (N)	Exhaustive Cross-Encoder Latency	Two-Stage Funnel (Dense + Top-10 CE)	Speedup Factor
This Experiment (GDPR)	444	**18.2 seconds**	**0.44 seconds**	**41.4x**
Enterprise Knowledge Base (Est.)	10,000	**410.3 seconds (~6.8 min)**	**~0.45 seconds (with ANN)**	**~910x**
Web / Legal Corpus (Est.)	1,000,000	**41,032.8 seconds (~11.4 hours)**	**~0.46 seconds (with ANN)**	**~89,000x**

Note: the 10k and 1M rows are theoretical projections that assume sub-linear ANN candidate retrieval for Stage 1. They were not measured.

4.3 Conclusion: The Two-Stage Funnel

Exhaustive cross-encoder retrieval is impractical for real-time interactive RAG: even this single 444-chunk document takes 18.2 s per query. The two-stage funnel is a practical engineering compromise:

- **Stage 1 (Bi-Encoder):** casts a wide net and narrows the corpus down to K = 10 candidates (20.4 ms measured here; with an ANN index this is expected to grow only sub-linearly with N).
- **Stage 2 (Cross-Encoder):** spends compute only on those K = 10 candidate pairs, a fixed cost of about 424 ms that does not depend on N.

This gives the precision of joint self-attention for a small, fixed re-ranking cost. Its limit is that Stage 2 can only be as good as the candidates Stage 1 returns (Section 2.3).

5. Experimental Setup & Limitations

- **Corpus & Chunking:** Official GDPR PDF (88 pages) split into 444 sentence-aligned chunks (target ~140 tokens, zero overlap).
- **Generation Settings:** Google Gemini `gemini-3.1-flash-lite`, temperature 0.0, with a prompt that strictly requires normative keyword retention (`SHALL`, `SHALL NOT`) and passage citations (`[C...]`).
- **Hardware:** Intel Core CPU laptop; models evaluated locally on CPU.
- **Limitations:** 16 hand-crafted queries give descriptive stress testing rather than statistical significance; single regulatory domain; the bi-encoder Stage 1 ceiling limits the maximum recall re-ranking can reach.